

CompTIA A+ Core1+ Playbook

IT Labs

Prepared by Garrick White

Compare Wireless Network Technologies

Assisted Lab: Compare Wireless Network Technologies

Scenario

You are working as an IT technician for the managed service provider (MSP) 515support. One of 515support's clients is the company Structureality, a virtual architecture design company who make digital models of building and furniture for virtual worlds. 515support is designing a network that will allow employees to connect to a LAN and also allow guests to connect to the Internet via a captive portal. This network is being modelled as a GNS3 project prior to deployment.

To complete this lab, make sure that the GNS3 model is fully configured and test that it establishes the expected connectivity.

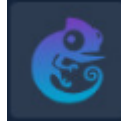
Objectives

This activity is designed to test your understanding of and ability to apply content examples in the following CompTIA A+ Core 1 objective:

- 2.3 Compare and contrast protocols for wireless networking.

Verify network topology

Open the GNS3 project and identify the devices used to implement the network topology and verify that they are connected correctly.



1. From the dock, select the **GNS3** icon .
2. Select **Recent projects >> 220-1101-wifi.gns3**.
3. In the main window, identify the following network appliances:
 - Router: this appliance has interfaces in three distinct zones (Internet, LAN, and Guest). The router can forward traffic between these zones. It can be set up with rules to prevent some types of connection. For example, hosts in the Guest zone are allowed to access the Internet, but are not allowed to access the LAN.
 - Switch: this appliance implements connectivity within the LAN zone. It allows PC1 to establish a wired Ethernet connection to the local network and also provides a wired connection to an access point (AP1). Wireless clients can then join the LAN via this AP.
 - AP1: this is one of the access points. It allows wireless stations to connect to the local network segment.
 - AP2: this is the other access point, hosting the guest zone. It is connected directly to the router.

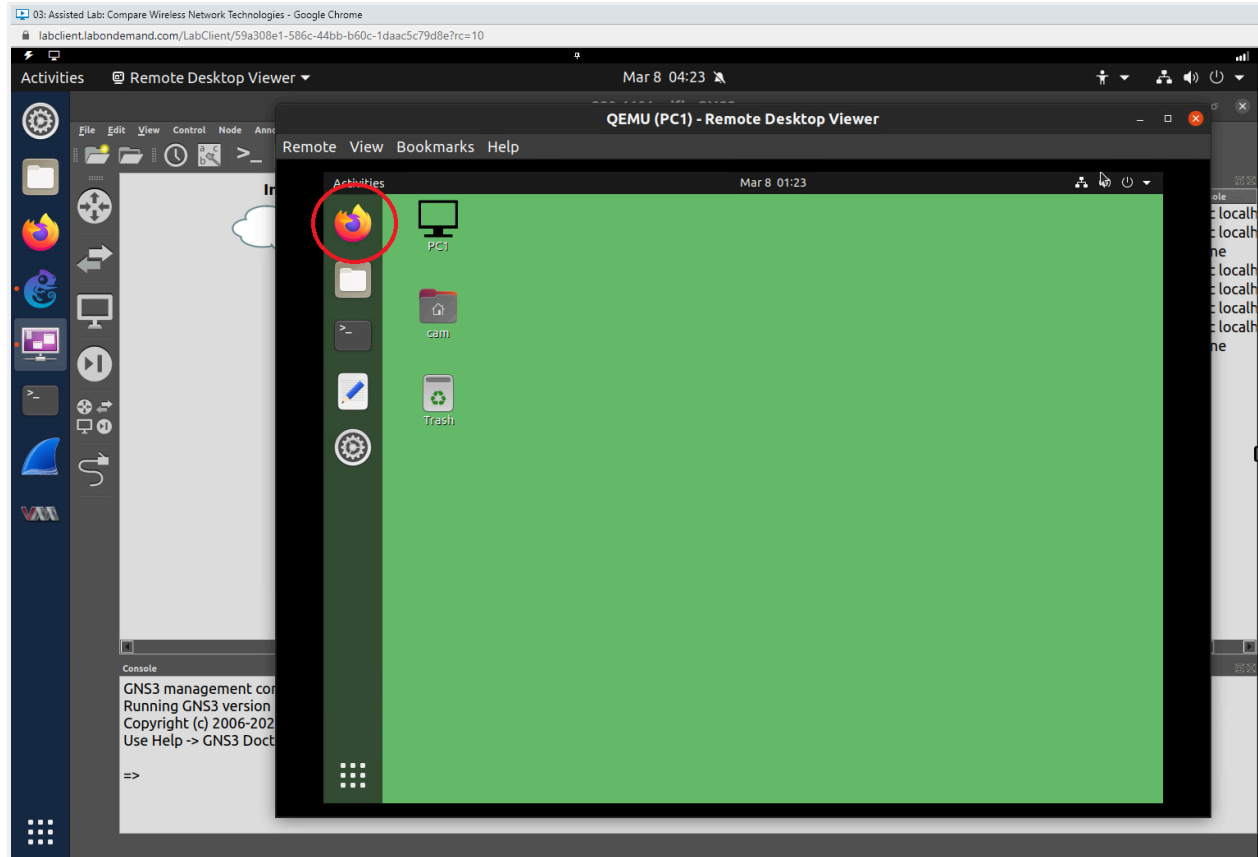
The router must be started and allowed to fully initialize before the other nodes.

4. Right-click the **Router** node and select **Start**.
5. Wait for half a minute and then click the **Check** button to verify that router services are started - if there is an error, click the button again to re-test:
6. When the router is identified as online, on the main toolbar, select



Start/Resume all nodes . Select **Yes** to confirm the prompt.

7. Right-click the **PC1** node and select **Console**.
PC1 is a nested virtual machine (VM) running Ubuntu. The console for this VM uses the graphical Remote Desktop Viewer rather than a simple terminal. Use the remote viewer window's title bar and the desktop icon to identify which nested VM you are operating. The VM desktop uses a different color scheme to distinguish it from the NSIM desktop.



When controlling a nested VM, use the dock and desktop within the remote viewer window.

8. In the **PC1** remote viewer window, from PC1's dock, open the **Firefox** web



browser .

9. In Firefox's address bar, type <http://10.1.0.1> and press **ENTER** to open the router's management interface.
10. Type **root** in the Username field and **Pa\$\$w0rd** in the Password field, then select the **Login** button.
11. From the menu on the left side of the browser window, select **Services > DHCPv4 > Leases**. Use the status page to record the IP addresses of AP1 and AP2.
 - AP1 10.1.0.2
 - AP2 10.254.0.2

Make sure you record these values accurately. They will be used later in the lab.

12. In Firefox's address bar, type the URL <https://comptia.org> and press **ENTER**.

CompTIA's home page should open, confirming that the Internet connection via the router is working.

Most sites now use HTTP Secure (HTTPS). This is essential for protecting information, but adds a complicating factor if you just want to verify basic connectivity and name resolution.

13. Close the browser window. Leave the PC1 remote viewer window open.

Connect LAPTOP1

The existing topology contains a LAPTOP1 end system that is supposed to be connected to the local network via the access point AP1. Ensure that LAPTOP1 is connected to the employees network.

1. Click **Deploy** to set the physical location of the emulated wireless devices used in this lab:

This simulates setting the range that any given wireless device is from another wireless device.

2. From the NSIM VM dock, select the **GNS3** icon. Right-click the **LAPTOP1** node and select **Console**.
3. In the **LAPTOP1** remote viewer window, click the **System** menu



from the top bar. Note that no connection is present.

4. Select **Wi-Fi Not Connected > Select Network**.
5. Select **employees** and click the **Connect** button.
6. Type **architects** in the password field and press **ENTER**.
7. Verify that the Wi-Fi icon in the top bar shows a solid connection



8. Open Firefox and browse <https://comptia.org> to verify that there is an Internet connection.
9. Select the **Score** button to validate this task:
10. Close the LAPTOP1 remote viewer window.

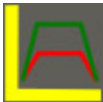
Connect LAPTOP2

The LAPTOP2-Guest node should be connected to the guest wireless LAN (WLAN) configured on AP2. Ensure that LAPTOP2-Guest is connected to the guest network.

1. From the NSIM dock, select the **GNS3** icon. Right-click the **LAPTOP2-Guest** node and select **Console**.
2. In the **LAPTOP2-Guest** remote viewer window, click the **System** menu  from the top bar. Note that no connection is present.
3. Select **Wi-Fi Not Connected > Select Network**.
4. Select **guest** and click the **Connect** button.
5. Type **structureality** in the password field and select **Connect**.
6. Verify that the Wi-Fi icon in the top bar shows a solid connection. This might take a few seconds.
7. In the **LAPTOP2-Guest** remote viewer window, open the **Firefox** web browser. Note the banner message on Firefox. Select **Open network login page**.
If there is a banner but not an **Open network login page** button, close the browser and open it again.
This is an example of a **captive portal**. A company can use this to authenticate guest users and to display terms and conditions. This particular portal does not require authentication, but guests must accept the terms before access to the web is granted.
8. A new web page will open in Firefox with the terms and conditions of Structureality's guest network. Scroll to the bottom of the page and select the **Sign in** button.
9. Check that you can use **Firefox** to browse <https://comptia.org>.
10. Select the **Score** button to validate this task:
11. Close the LAPTOP2-Guest remote viewer window.

Verify configuration

You might have noticed that the Internet connection from one or both LAPTOP VMs was slow. Use the [LinSSID Wi-Fi analyzer](#) to check the network for a possible misconfiguration.

1. From the NSIM VM dock, select the **GNS3** icon. Right-click the **LAPTOP1** node, and select **Console**.
2. In the **LAPTOP1** remote viewer window, from the dock, select **LinSSID** . When prompted, enter **Pa\$\$w0rd** in the password field and select

Authenticate.

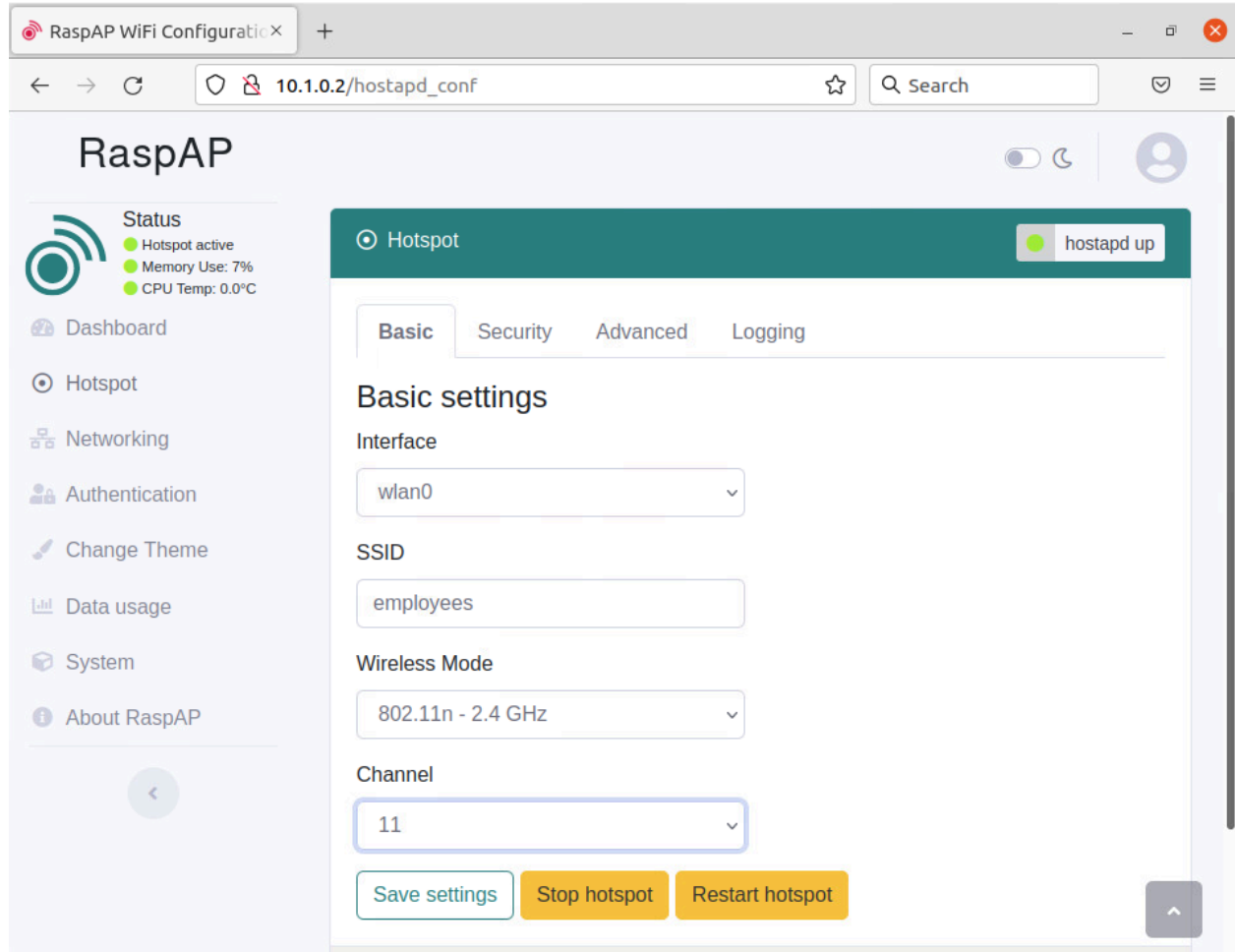
A packet analyzer requires root or administrative permissions to operate the wireless adapter in scanning/monitor mode.

3. Select the **Play** button to start the scanner.
After a few seconds, the analyzer should report the presence of the employees and guest WLANs.
4. Maximize the LinSSID window and, if necessary, use the scroll bar to view all the information reported about the networks, including the protocol type.
5. Select the **2.4 GHz Channels** tab to view the channel graph.
6. Which of the following factors is MOST likely to affect the performance and reliability of the two networks?
Both APs are configured to use channel 1
7. Select the **Pause** button to stop scanning.

Remediate configuration

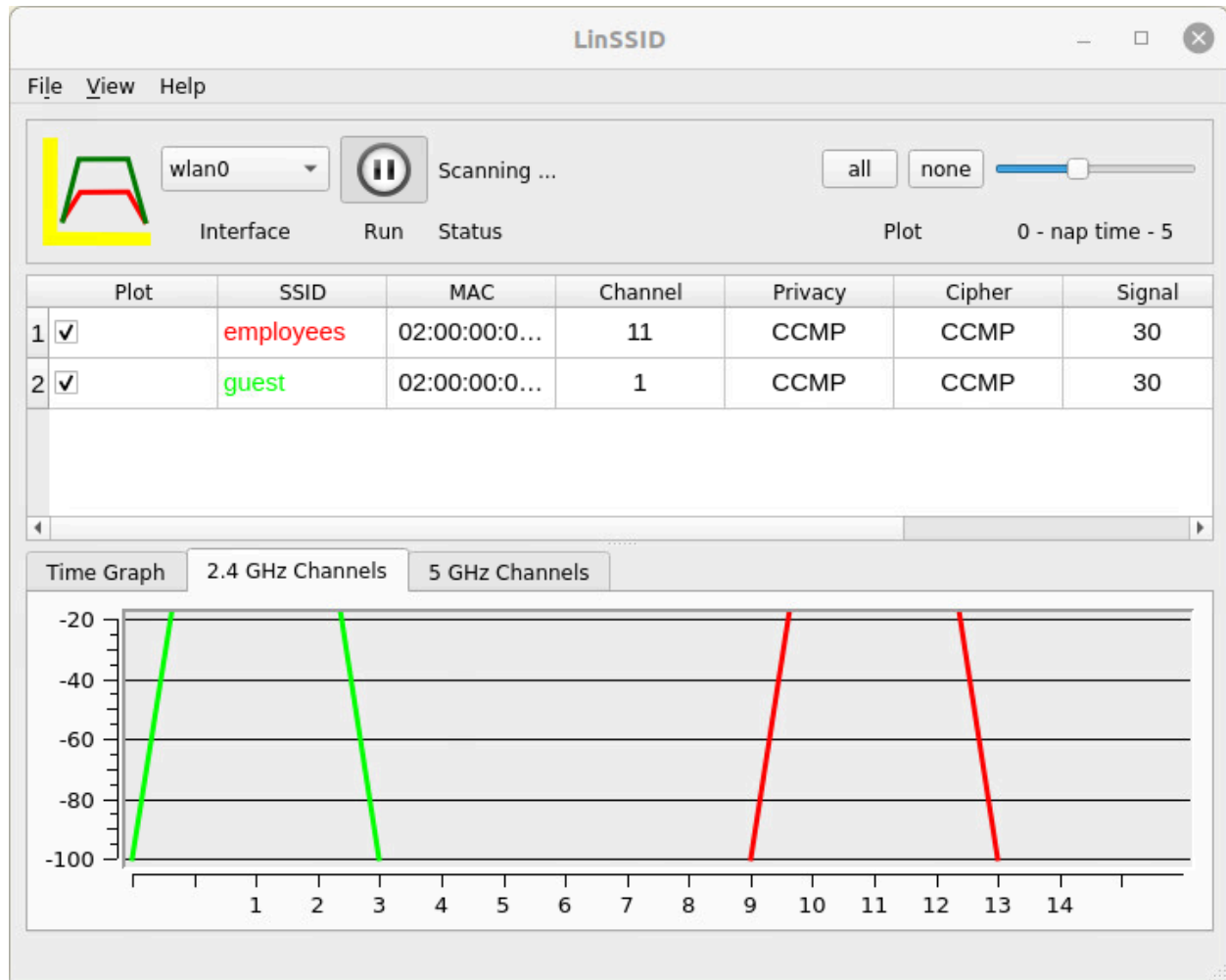
As they are in the same geographical location and both have strong signals, the wireless LANs should not be configured to use the same channel. This will cause unnecessary interference. Reconfigure the employees WLAN to use a clear channel.

1. If one access point is configured to use channel 1, which of the following channels is the best channel for the other AP to use?
1
2
6
(11) correct
36
40
2. From the NSIM VM dock, select the **GNS3** icon. Right-click the **PC1** node, and select **Console**.
3. In the **PC1** remote viewer window tab, select the **Firefox Web Browser** app from the dock and open <http://10.1.0.2>.
If there is no connection, check that you recorded the IP address of AP1 correctly.
4. When prompted, sign in by typing **admin** in the Username field and the Password **Pa\$\$w0rd** in the password field and select the **Sign in** button.
5. In the **RaspAP** admin page, select **Hotspot**.
6. From the **Channel** list box, select **11**. Select **Save Settings**.



Use the RaspAP admin page to change the channel ID.

7. Select **Restart Hotspot** to apply the changes.
8. In the Remote Desktop Viewer window, select the **QEMU (LAPTOP1)** tab.
9. From the **System** menu, select **Employees > Turn off** and then **Wi-Fi Off > Turn On** to toggle the adapter off and on and detect the new channel.
10. Select the **Play** button to start the scanner.
11. View the LinSSID output to verify that it reports the new channel layout.



Check that the nearby networks are configured to use non-overlapping channels to minimize interference.

If LinSSID does not update, restart the app, authenticating with `Pa$$w0rd`.

Alternatively, you can use the `wavemon` tool from the terminal.